SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I ________BAIRAGANI DINESH 192372189__________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Question 1: Intelligent Agents in a Smart Home Automation System

1. Types of Intelligent Agents and Their Application

Simple Reflex Agent

- **Working Principle:** Operates strictly on condition-action rules (IF condition $\Longrightarrow$ THEN action) based entirely on the current percept, completely ignoring percept history.
- **Function in Smart Home:**

- **Lighting:** IF motion sensor detects movement in a room, THEN turn on lights.
- **Temperature:** IF temperature exceeds 24°C, THEN turn on the air conditioner.
- **SecUrity:** IF a window contact sensor breaks while armed, THEN trigger the alarm.

Model-Based Reflex Agent:-

- **Working Principle:** Maintains an internal state (a model of how the world works and how actions change it) to handle partial observability by combining current percepts with historical background context.
- **Function in Smart Home:**
  - Tracks occupancy state over time. For example, if motion was detected 2 minutes ago and a door hasn't opened, the internal state maintains "Room Occupied" even if no current motion is detected, preventing lights from turning off unexpectedly.

Goal-Based Agent:-

- **Working Principle:** Combines internal state information with explicit goal descriptions to plan sequence of actions that transition the system from its current state to a target state.
- **Function in Smart Home:**
  - **Goals:** "Maintain temperature at 22°C by 6:00 PM," "Ensure all entry points are secured by 10:00 PM," or "Achieve zero standby power consumption during vacation."
  - **Decision Making:** Instead of reacting purely to immediate signals, it evaluates sequences of actions (e.g., pre-cooling the house at off-peak rates to meet the 6:00 PM temperature goal).

Utility-Based Agent:-

- **Working Principle:** Evaluates how desirable competing states are using a utility function. It maps states to a real number measuring performance trade-offs under uncertainty.

2. Justification of the Most Effective Approach

A **Learning Utility-Based Hybrid Architecture** (a Learning Agent combined with a Utility-Based core) is the most effective approach for this system.

Justification:

1. **Trade-off Optimization:** Comfort, energy efficiency, and security frequently conflict. A purely goal-based agent treats goals as binary (achieved or not), whereas a utility-based core quantifies trade-offs seamlessly.
2. **Preference Adaptation:** Incorporating a **learning element** enables the system to construct and continuously adjust user preference profiles over time from interaction logs (e.g., manual thermostat overrides), moving beyond static hardcoded rules.

Question 2: Robot Maze Navigation using Uninformed Search

1. Uninformed Search Strategies in Unknown Mazes:-

Uniform Cost Search (UCS):-

- **Mechanism:** Expands nodes according to the path cost $g(n)$ from the root. It expands the unvisited node with the lowest cumulative path cost first using a priority queue.
- **Maze Application:** If different terrain in the maze incurs different costs (e.g., smooth tiles vs. rough rubble vs. battery consumption turning corners), UCS guarantees finding the lowest-cost path from start to exit. If all step costs are equal (1), UCS behaves identically to Breadth-First Search (BFS).

Iterative Deepening Search (IDS):-

- **Mechanism:** Combines the depth-first search strategy with incremental depth limits ($d = 0,1,2, \dots$). It performs Depth-First Search (DFS) up to a maximum depth d, increasing d iteratively until a goal state is reached.

2. Time and Space Complexity Analysis:-

Let b be the branching factor (average number of open directions at a maze junction) and d be the depth of the optimal solution. Let C^* be the cost of the optimal solution and ϵ be the minimum step cost.

Search Algorithm	Time Complexity	Space Complexity	Optimality	Completeness
Uniform Cost Search (UCS)	$O\left(b^{1+\lfloor C^*/\epsilon \rfloor}\right)$	$O\left(b^{1+\lfloor C^*/\epsilon \rfloor}\right)$	**Yes** (if step costs $\geq \epsilon > 0$)	**Yes**
Iterative Deepening Search (IDS)	$O(b^d)$	$O(b \cdot d)$	**Yes** (if step costs are equal)	**Yes**

Comparison:

- **Space Complexity:** IDS is far superior in memory efficiency ($O(b \cdot d)$ vs $O(b^d)$). In large unknown mazes, UCS can rapidly exhaust system memory storing all frontier nodes.

3. Agent Design and Decision-Making

- **Model-Based Reflex / Goal-Based Structure:** In an unknown maze, the robot lacks initial global map details. A model-based agent updates its internal map state (occupancy grid or topological graph) as sensors (e.g., LiDAR/ultrasonic) observe local walls.
- **Impact of Agent Design:**
    - A **Simple Reflex Agent** would move randomly or follow a wall (e.g., right-hand rule), failing completely in mazes with loops or detached inner walls.
    - A **Goal-Based Agent** uses its mapped internal state to run search algorithms (UCS/IDS) repeatedly to find valid paths toward the exit as new walls/passages are discovered.

4. Recommended Combination and Justification:-

Recommended Strategy: A **Model-Based Goal Agent utilizing Iterative Deepening Search (IDS)** (or **Uniform Cost Search** if step costs vary widely and memory permits).

Justification:

1. **Memory Constraints:** Embedded robotics operate under strict memory limits. IDS guarantees finding the shortest topological exit path with linear space consumption ($O(b \cdot d)$), preventing memory overflow during large maze

exploration.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

		flow and coherence			
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand